

PRE LESSON

- Ask students to check their mass in kg
(OR have bathroom scales in lab for them to find it out, but be conscious this may be a sensitive issue for some).

Pressure Between Solids

1. Define a moment

The turning effect of a force about a pivot

2. What is the unit for moments?

Newton meter

3. Calculate the moment of a lever with 500N of force acting on a pivot at 15m distance

$$M = f \times d$$

$$M = 500 \times 15 = 7500 \text{ Newton meters}$$

4. Describe the properties of a solid

Definite shape and volume, does not flow or change shape under pressure

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.



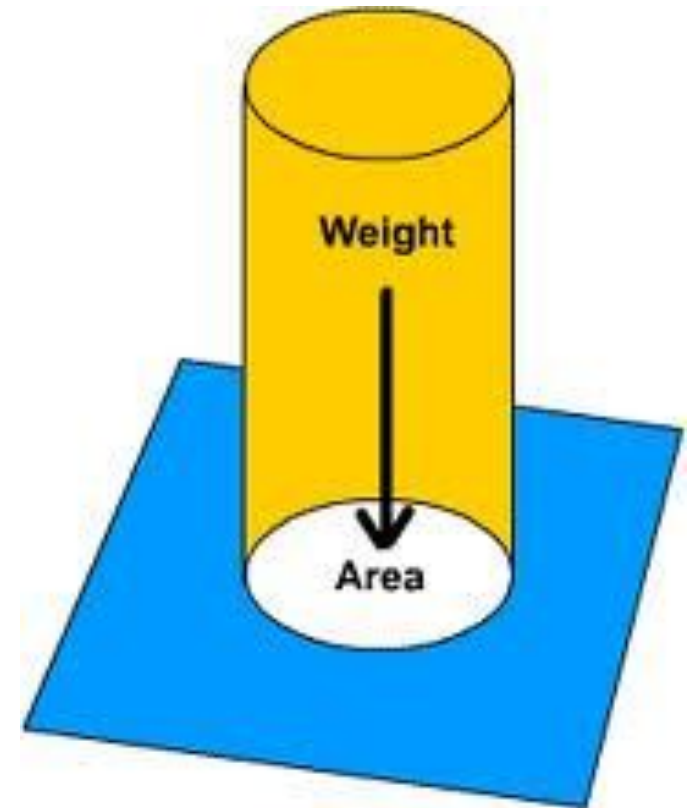
Which has the
most pressure
on the thumb? A
or B?



Pressure

- Pressure is how much force is applied to a certain area.
- Therefore we calculate pressure as:

$$\text{Pressure} = \frac{\text{Force (N)}}{\text{Area (cm}^2\text{)}}$$



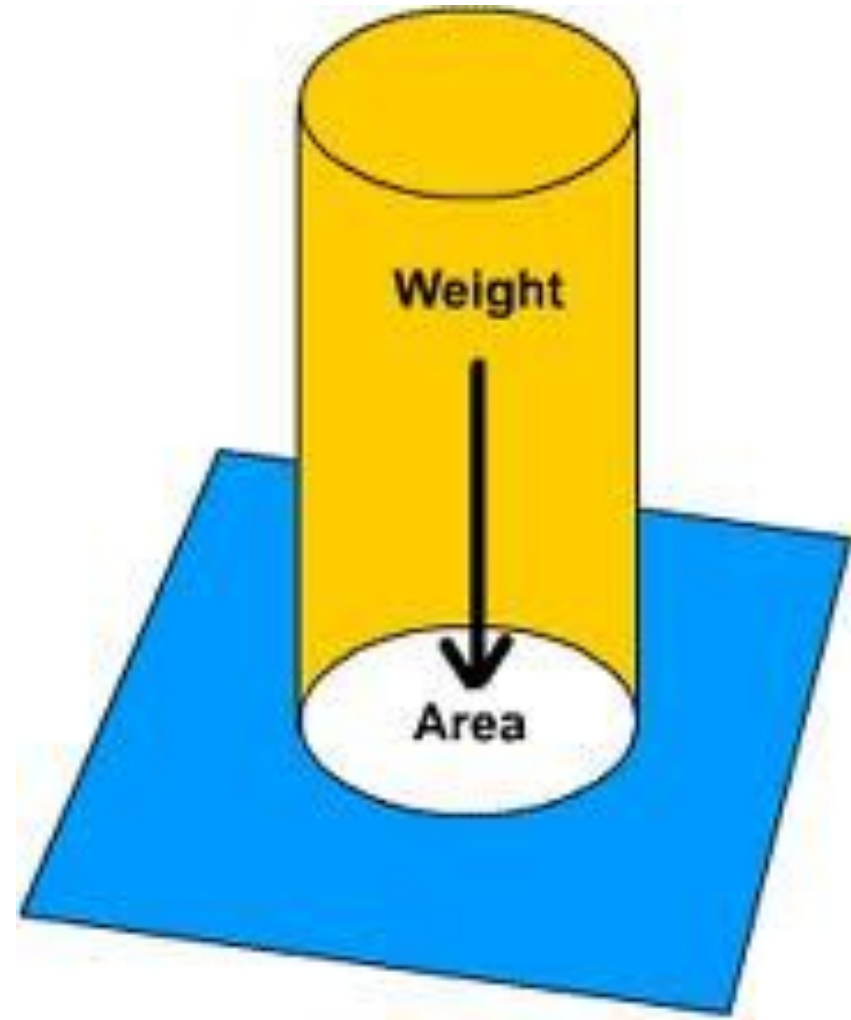
Pressure

$$\text{Pressure} = \frac{\text{Force (N)}}{\text{Area (cm}^2\text{)}}$$

So what are the units of Pressure?

N/cm² From the formula above.

N/m² and Pascals are also accepted units



Calculating pressure

Photocredit: Freepik

- A skier has a force of 75N and his skis cover an area of 30cm², calculate the pressure of the skier.



$$\text{Pressure (N/cm}^2\text{)} = \frac{\text{Force (N)}}{\text{Area (cm}^2\text{)}}$$

$$P = 75\text{N} / 30\text{cm}^2$$

$$P = 2.5 \text{ N/cm}^2$$

Calculating pressure

- A drawing pin is pushed with a force of 50N into an area of 0.5cm^2 . Calculate the pressure.



$$P = 50\text{N} / 0.5\text{cm}^2$$

$$P = 100 \text{ N/cm}^2$$

$$\text{Pressure (N/cm}^2\text{)} = \frac{\text{Force (N)}}{\text{Area (cm}^2\text{)}}$$

Scenarios

Explain to the person next to you...

Why does it REALLY hurt a lot more if you drop stool foot on my foot while you're sat on it, more than if you trod on my toe.

Explain to the person next to you...



Calculating pressure

In this investigation, you will investigate how the pressure that a person exerts on the floor varies.

Work in groups of two or three.

You will need:

- a person who is willing to be weighed, or who knows their body mass
- a bathroom scale, if body mass is not known
- sheets of squared paper
- pencil

Method

- 1 Measure or record the body mass of the person.
- 2 Convert the mass to weight using:
$$\text{weight in N} = \text{mass in kg} \times \text{strength of gravity in N/kg}$$

Take the strength of gravity as 10 N/kg.
- 3 Ask the person to place one foot on a piece of squared paper. They can do this while wearing shoes.
- 4 Use the pencil to draw around the foot of the person.

Questions

- 1 Use the shape of the outline on the squared paper to work out the area of the foot in cm^2 .
- 2 Calculate the area of both feet.
- 3 Calculate the pressure that the person exerts on the ground when they are:
 - a standing equally on both feet
 - b standing on one foot.
- 4 Explain the difference in your answers to **3a** and **3b**.
- 5 Now ask the same person to put the front of one foot on squared paper as if they were standing on their toes.

Use the same method as above to work out the area of the front of the foot in cm^2 .
- 6 Calculate the pressure exerted by the person on the ground when standing on the front of one foot.
- 7 A person can be supported by the front of one foot during some everyday activities. Give an example of such an activity.

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Continued

- 8
 - a Describe how you could work out the area in contact with the ground when the person is lying down.
 - b
 - i Predict how the pressure would change when the person was lying down compared to standing.
 - ii Explain your answer.

Self-assessment

- 1 Decide how confident you are about:
 - a how force affects pressure
 - b how area affects pressure.
- 2 Decide how confident you are about:
 - a calculating pressure when you know the force and the area
 - b working out the unit of pressure using the units of force and area.

What is your pressure?

1. Trace your foot on the graph paper to find the area of your foot.

2. What is your WEIGHT!? (think carefully)

Convert the mass to weight using:

$$\text{weight in N} = \text{mass in kg} \times \text{strength of gravity in N/kg}$$

3. NOW WORK OUT YOUR PRESSURE:

- Standing on 2 feet
- Standing on 1 foot

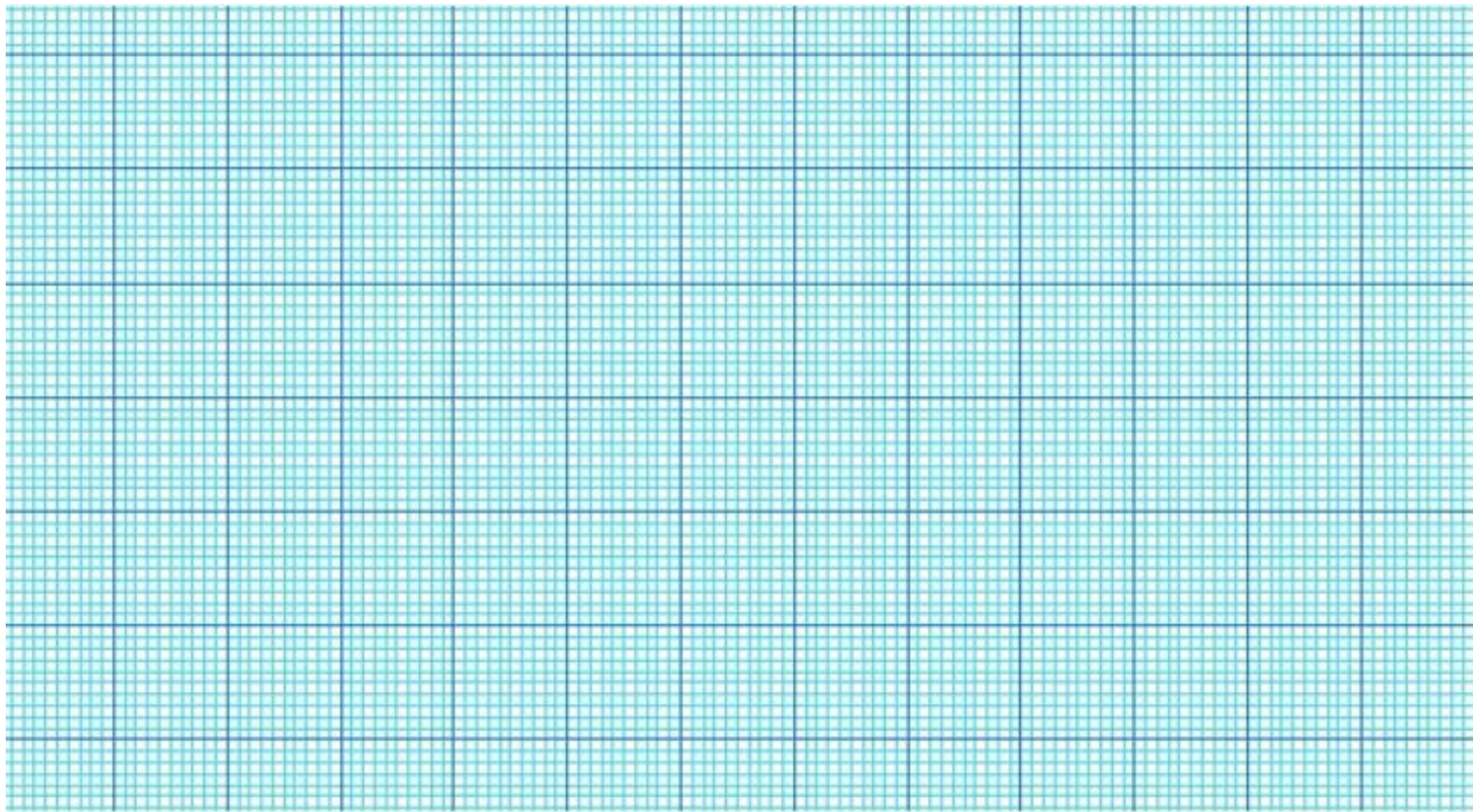
1 full box = 1cm²

	Area cm ²	Pressure, N/cm ²
1 foot		
2 feet		

Top Tips

- Count large squares only

To convert Kg to Newtons $\times 10$



What affects pressure?

Force

- $\uparrow$ Force = $\uparrow$ Pressure
- $\downarrow$ Force = $\downarrow$ Pressure

Surface Area

- $\uparrow$ Surface Area = $\downarrow$ Pressure
- $\downarrow$ Surface Area = $\uparrow$ Pressure

What affects pressure?

Force

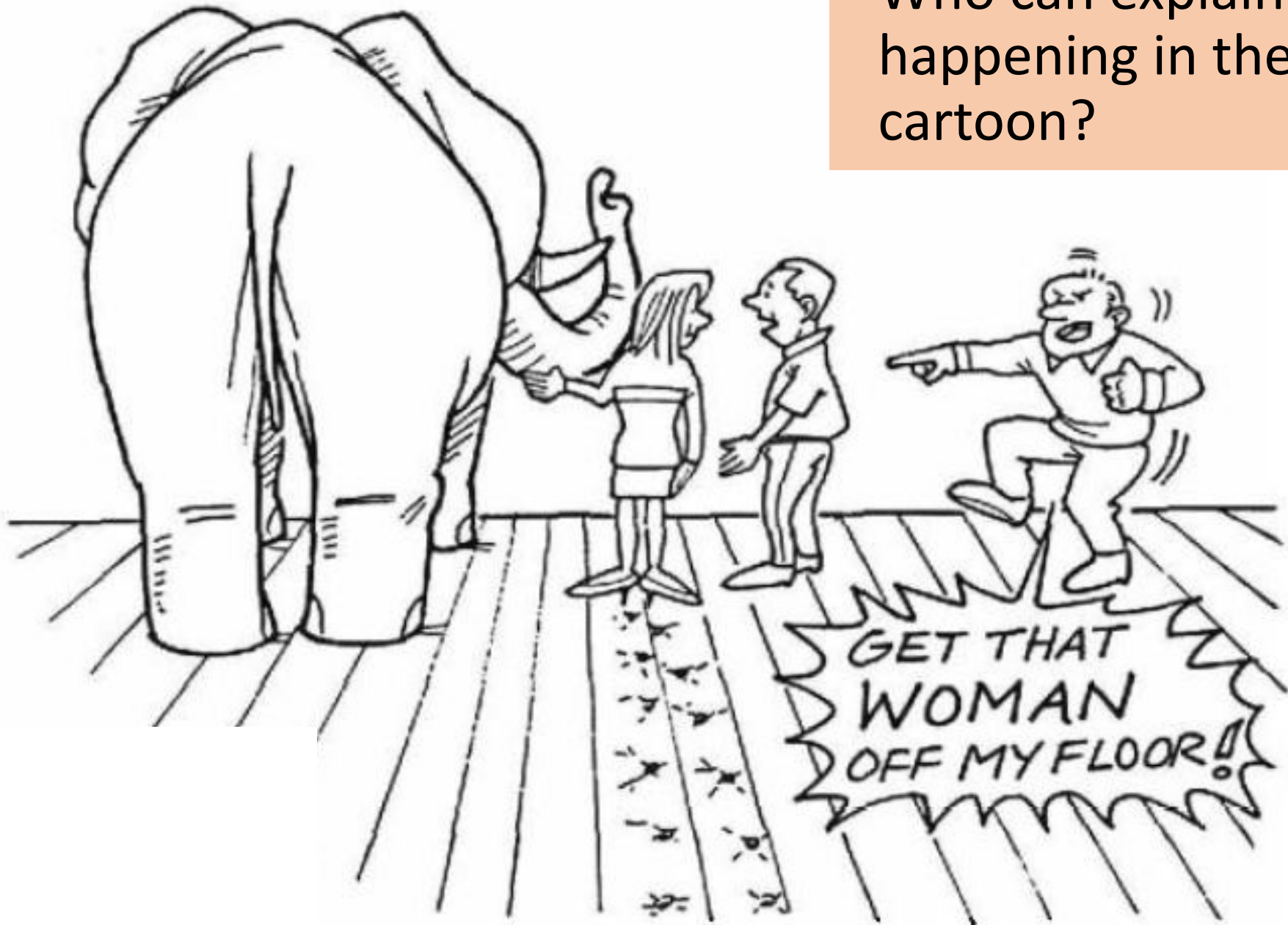
- $\uparrow \text{ Force} = \uparrow \text{ Pressure}$
- $\downarrow \text{ Force} = \downarrow \text{ Pressure}$

Surface Area

- $\uparrow \text{ Surface Area} = \downarrow \text{ Pressure}$
- $\downarrow \text{ Surface Area} = \uparrow \text{ Pressure}$

But is your pressure greater than an elephant?!

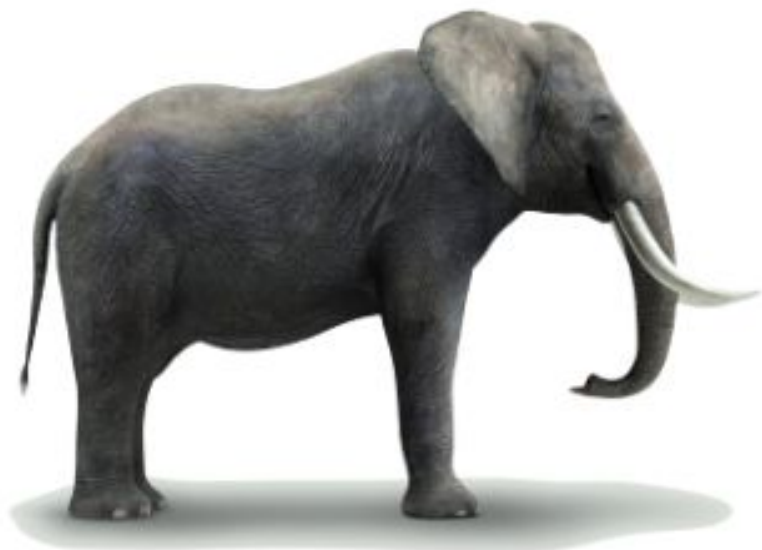
- Who can explain what's happening in the cartoon?



Worked example

Question

An elephant stands on four feet.



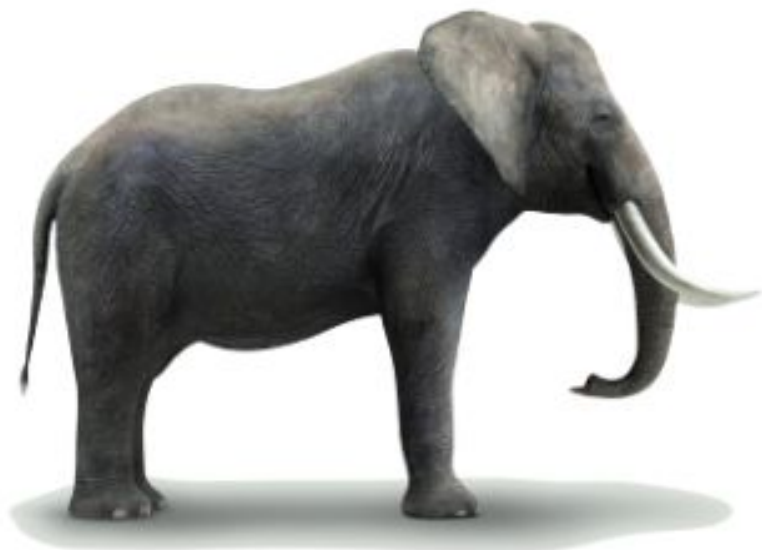
The weight of the elephant is 50 000 N.
The total area of all four feet is 0.4 m^2 .

What is the total pressure that the elephant exerts on the ground?

Worked example

Question

An elephant stands on four feet.



The weight of the elephant is 50 000 N.
The total area of all four feet is 0.4 m².

What is the total pressure that the elephant exerts on the ground?

Answer

$$\begin{aligned}\text{pressure} &= \frac{\text{force}}{\text{area}} \\ &= \frac{50\,000}{0.4} \\ &= 125\,000 \text{ N/m}^2\end{aligned}$$

The unit of pressure here is N/m² because the area is given in m².

Question

The total weight of a bicycle and rider is 1000 N.

The bicycle has two tyres in contact with the ground. The weight is supported equally on both tyres.

The area of each tyre in contact with the ground is 5 cm^2 .

What is the pressure that each tyre exerts on the ground?

Question

The total weight of a bicycle and rider is 1000 N.

The bicycle has two tyres in contact with the ground. The weight is supported equally on both tyres.

The area of each tyre in contact with the ground is 5 cm².

What is the pressure that each tyre exerts on the ground?

Answer

$$\begin{aligned}\text{weight on each tyre} &= \frac{1000}{2} \\ &= 500 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{pressure} &= \frac{\text{force}}{\text{area}} \\ &= \frac{500}{5} \\ &= 100 \text{ N/cm}^2\end{aligned}$$

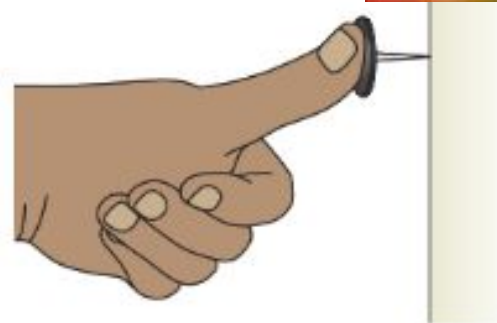
The unit of pressure here is N/cm² because the area of the tyres is given in cm².

Questions

- 1
 - a Write down the equation that links pressure, force and area.
 - b Use the equation to explain why:
 - i a nail that goes into wood has a sharp point
 - ii shoes for walking on snow are wide and flat.
- 2 Write down the unit of pressure when force is in newtons and area is in:
 - a m^2
 - b cm^2
 - c mm^2
- 3 A box has a weight of 60 N. The area of the box in contact with the ground is 0.5 m^2 .
Calculate the pressure that the box exerts on the ground.
Show your working and give your answer in N/m^2 .
- 4 A car has a weight of 8000 N. The car is supported by four tyres. The weight on each tyre is equal.
The area of **one** tyre in contact with the ground is 150 cm^2 .
Calculate the pressure that **one** tyre exerts on the ground.
Show your working and give your answer in N/cm^2 .
- 5 A thumb tack has an area of 0.5 mm^2 in contact with a wall.
A pressure of 40 N/mm^2 is needed for the thumb tack to go into the wall.
Calculate the force needed to push the thumb tack into the wall.
Show your working and give the unit with your answer.

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Remember to show your working and include the units!



Answers

Mark your work using a green pen!

1 a $\text{pressure} = \frac{\text{force}}{\text{area}}$

b i To make the area as small as possible, so the pressure is as large as possible.

ii To make the area as large as possible, so the pressure is as small as possible.

2 a N/m^2

b N/cm^2

c N/mm^2

3
$$\begin{aligned}\text{pressure} &= \frac{\text{force}}{\text{area}} \\ &= \frac{60}{0.5} \\ &= 120 \text{ N/m}^2\end{aligned}$$

4 Total weight = 8000 N

$$\begin{aligned}\text{weight on each tyre} &= \frac{8000}{4} \\ &= 2000 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{pressure on each tyre} &= \frac{\text{force}}{\text{area}} \\ &= \frac{2000}{150} \\ &= 13.3 \text{ N/cm}^2\end{aligned}$$

5 $\text{pressure} = \frac{\text{force}}{\text{area}}$

$$\begin{aligned}\text{force} &= \text{pressure} \times \text{area} \\ &= 40 \times 0.5 \\ &= 20 \text{ N}\end{aligned}$$